

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CL)

November – December 2011

CL 204 Engineering Geology

Date: 02.12.2011, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Define Earthquake. Discuss in detail tectonic and non-tectonic causes of earthquake. [07]

Q - 2 (a) What are the objectives of engineering geology? What are the applications of geology in water resources development and town planning? [07]

(b) Write short notes on: [07]

(i) Physical weathering process on rock mass

(ii) Wind erosion

OR

Q - 2 (a) Write briefly note on argillaceous, arenaceous and rudaceous types of sedimentary rocks along with any two suitable examples of each rock. [07]

(b) Write short notes on: [07]

(i) Chemical weathering process on rock mass

(ii) Desert advancement and its control

Q - 3 (a) Describe various structures developed in igneous rocks. Add your note with subdivision of igneous rocks based on their occurrence. [07]

(b) Write short notes on following: [07]

(i) Colour and luster property of mineral

(ii) Metamorphic effects on various rocks

OR

Q - 3 (a) Describe various structures formed in sedimentary rocks. Add your note with various facies in sedimentary rocks. [07]

(b) Write short notes on following: [07]

(i) Cleavage and fracture in mineral

(ii) Textures in igneous rocks

SECTION – II

- Q - 4 (a)** Define 'mass movement'. Write a note on external factors causing mass movement. [04]
- (b)** Describe confined aquifer system and artesian well condition. [03]
- Q - 5 (a)** (i) Write a note on various causes of fold. [05]
- (ii) How fault can be recognized in the field? [02]
- (b)** Write short notes on following: [07]
- (i) Occurrence of joints in rocks
- (ii) Geological problems of reservoir

OR

- Q - 5 (a)** (i) Write a note on various plate boundaries and associated fault & fold structures. [05]
- (ii) Enlist separately igneous, sedimentary and metamorphic minerals. [02]
- (b)** Attempt followings: [07]
- (i) Differentiate between various major unconformities
- (ii) Classification of joints
- Q - 6 (a)** Explain following: [07]
- (i) Forces on dam due to weight of dam, uplift and water pressure
- (ii) Geological profile in tunneling operations and its importance
- (b)** Discuss in detail followings: [07]
- (i) How ground water conditions can influence tunneling work?
- (ii) How porosity, permeability and specific yield of rock mass governs the water movement in the rock formations?

OR

- Q - 6 (a)** Give answers for following: [07]
- (i) How geological maps can be helpful for civil engineering structures?
- (ii) Whether our residential structures can withstand the earthquake of magnitude of M:6.9 ? Comment on the statement giving reasons.
- (b)** Write short notes on: [07]
- (i) Study of geological characters for dam site selection
- (ii) Influence of geological factors on design of tunnel
